

Herald



Herald — Status

Field	Value
Revision	15
Created	2026-05-20
Last modified	2026-05-22
Status	active
Status summary	r15 captures Wave 4b (TOON application/toon content negotiation). Tag v0.3.0 issued. Every <code>/v1/*</code> business route on every serving flavor now honors <code>Accept: application/toon</code> → real TOON-encoded body (via <code>digital.vasic.toon+toon-format/toon-go</code> upstream wire-up replacing the prior sentinel-error scaffold from the §107 PASS-bluff revert). JSON remains the honest fallback. <code>POST /v1/events</code> accepts both TOON and JSON request bodies (TOON transcoded to JSON before the Runner's EventParser; Runner stays codec-agnostic). <code>HERALD_DEFAULT_RESPONSE_CODEC=json</code> operator opt-out. <code>TOONMiddleware</code> auto-wired into <code>cli.RunServe</code> — every flavor binary (<code>pherald/cherald/sherald/bherald/rherald/iherald/scherald</code>) inherits for free. Real compression evidence: <code>sherald /v1/safety_state TOON=215B</code> vs <code>JSON=226B</code> (~5% on a small payload; larger tabular payloads see 30-60% per upstream measurements). 7 new e2e invariants E56-E62 (E62 SKIP-with-reason — encoder-failure fallback unit-tested in <code>commons/cli/toon_test.go</code>). New <code>tests/test_wave4b_mutation_meta.sh</code> (5/5 PASS — M1 strip <code>TOONMiddleware</code> → E57 FAIL; M2 swap <code>toon.Marshal</code> → <code>json.Marshal</code> (the EXACT 2026-05-17 PASS-bluff signature) → E60 FAIL — gate proves wire-byte-0 anti-bluff catches the recurrence; M3 no-op <code>transcodeRequestBody</code> → <code>pherald</code> unit test FAIL; working-tree quiescence guard (lesson from 72e81ab — script greps own working tree for residual mutation markers BEFORE declaring PASS); post-flight green). Workspace module count unchanged at 15 (no new module). Tag v0.3.0. Prior r14 captured Wave 4a (HTTP/3 + Brotli + Alt-Svc + TLS 1.3; tag v0.2.0). r13 closed HRD-016 (Wave 3b pherald Runner live). r12 closed HRD-011 (Telegram live). r10-r11: Wave 3a substrate.
Issues	HRD-008, HRD-015, HRD-018 (in_progress), HRD-019..HRD-027, HRD-029..HRD-056, HRD-081, HRD-085..HRD-090
Issues summary	see <code>Issues.md</code> — 47 open items spanning §42 constitution bindings (HRD-018..027), §43 command-catalogue (HRD-029..056), HRD-081 (containers podman-compose adapter refinement), HRD-085..090 (queue-repository stub-completion surface). No HRDs migrated in r15 (substrate upgrade only — Wave

Field	Value
	4b). Prior cycle: HRD-016 (pherald Runner) + HRD-011 (Telegram live) + HRD-028 + HRD-098 + HRD-099 closed in r10-r13.
Fixed	HRD-001..HRD-007, HRD-009, HRD-009b, HRD-010, HRD-011, HRD-012, HRD-013, HRD-014, HRD-016, HRD-017, HRD-028, HRD-080, HRD-092, HRD-093, HRD-094, HRD-095, HRD-096, HRD-097, HRD-098, HRD-099
Fixed summary	see <code>Fixed.md</code>
Continuation	QA workstream next: Constitutional addition for <code>docs/qa/</code> evidence mandate + <code>qaherald</code> Go binary driving full pherald ☐ Telegram round-trip automation with attached materials preserved under <code>docs/qa/<run-id>/</code>; final close-out is tag <code>v1.0.0-dev-0.0.1</code> (or canonical-named) release. Carry-over Wave 3c items (HRD-024 iherald paging, HRD-033 destructive-guard body, HRD-018..027 flavor bindings, the rest of the §43 command catalogue) remain queued. Each milestone followed by multi-mirror push + full anti-bluff verification.

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Project phase

Implementation-r1, Foundation complete + Wave 2 flavor scaffolds landed. Spec V3 r8 is active, the Go module foundation compiles, all three Foundation milestones (M1/M2/M3) plus the Wave 2 flavor-scaffold workstream have landed, and the §107 end-user-usability covenant is now explicit in Herald's three root docs + gate (added 2026-05-20). The remaining cycle is wiring live channel integrations (HRD-011, HRD-016, HRD-024, HRD-028, HRD-098), validating the §26.5 quickstart end-to-end on operator hardware, and implementing the 28 §43 command bodies still stubbed.

Specification

- **Active spec:** `docs/specs/mvp/specification.V3.md` (Revision 8, ~4600 lines after r8 Wave 2 capture).
- **Archived specs:** V1 + V2 in `docs/specs/mvp/archive/` (frozen).
- **Recent sections:** §37 Tracker-doc change events; §38 Workable-item announcement contract; §39 Message presentation + Herald Canonical Template; §40 Documentation + 15 named test challenges; §41 REST API surface (Gin Gonic); §42 Constitution-binding integration; §43 Constitution-derived flavor commands (27 entries → HRD-029..HRD-056) + §43.2 catalogue-check addendum (r8); §44 Foundation implementation contract + §44.M Wave 2 milestone capture (r8); §6.3 Branding extension with per-flavor fields (r8); §18.0 Wave 2 flavor-binary catalogue (r8).
- **Spec-change rule:** §23 + HERALD_CONSTITUTION §106 + I7 gate — green throughout.
- **End-user-usability covenant:** HERALD_CONSTITUTION §107 + I8 gate (added 2026-05-20) — green; paired §1.1 mutation meta-test in `tests/test_i8_usability_meta.sh` (5/5 PASS).

Implementation

Module	Status	Notes
commons (L0)	✅ landed	§11.0 types + Clock + UUIDv7 + DefaultBranding; tests pass.
commons_prefix	✅ landed	§8.2 algorithm + 5 test functions; tests pass.
commons_constitution (M1)	✅ landed	14 files, ~2.9k LOC — Evaluator + Registry + 12 emit helpers + BundleHash + ModeLadder + ConstitutionStore + MemoryBus + Runner + CloudEvents v1.0 adapter; all green under <code>go test -race</code> .
commons_messaging (L1)	partial	null:// adapter fully working; Telegram + Claude Code stubs in place (HRD-011/012 for live wiring). 9 SQL migrations embedded; pgx pool + RLS-enforcing WithTenantContext live (commit c8168ec); HRD-010 (commits 82ea82d..13cea69) wired queue (digital.vasic.background → pgxTaskRepository) + Redis (digital.vasic.cache) + the pherald migrate up/status/down/validate CLI. The §107 covenant proved its worth here: the E14 RLS falsifiability test caught (and the implementer fixed) a real production RLS-bypass bug where the bootstrap PG superuser silently bypassed <code>FORCE ROW LEVEL SECURITY</code>. HRD-085..090 opened to track the remaining 16 pgxTaskRepository methods returning <code>ErrUnsupported</code>.
commons_storage (L1)	✅ landed (M2 + HRD-010)	
commons_infra	✅ landed	QuickstartBoot + WithTenantContext + on-demand container orchestration via <code>containers/</code> .
commons/cli (L0)	✅ landed (Wave 2 / HRD-092)	Shared CLI scaffold: NewRootCmd + VersionCmd + ServeCmd (with Middleware hook) + StubCmd + healthz/readyz/metrics handlers. Consumed by pherald + 6 new flavor binaries. Catalogue-check no-match → vendored Herald-internal per §11.4.74.
pherald (CLI)	✅ partial (M3 + Wave 2 refactor)	Cobra root + version + serve (live HTTP on :24791, /v1/healthz, /v1/readyz, /v1/events, /metrics) now consumes cli.ServeCmd with RequestIDMiddleware hook (commits 1f81d69, 31562cf). send/doctor/migrate/subscriber/deadletter stubbed with HRD pointers. System Herald binary serving :24793 via cli.ServeCmd + commons_auth.GinMiddleware (JWT gate). /v1/safety_state now LIVE — backed by sherald/internal/safety (process-local Aggregator + 15s mem-sampler goroutine, lifecycle bound to SIGTERM-cancelled context). 5 §43 stubs (HRD-033/034/040/046/056) still return honest non-zero + HRD pointer.
sherald (CLI)	✅ live (Wave 3a / HRD-098)	Constitution Herald binary serving :24792 via cli.ServeCmd + commons_auth.GinMiddleware (JWT gate). /v1/compliance now LIVE —
cherald (CLI)	✅ live (Wave 3a / HRD-028)	

Module	Status	Notes
		backed by <code>cherald/internal/compliance</code> (paginated + filter-aware <code>ConstitutionStore</code> reads; tenant scope extracted from JWT claims). 11 §43 stubs still return honest non-zero + HRD pointer.
<code>commons_auth</code>	✅ landed (Wave 3a / HRD-099, renumbered from HRD-093)	JWT verifier — HMAC (HS256) + JWKS (RS256/ES256) hybrid; Redis-cached keys with in-memory fallback; Gin middleware + claims helpers. Catalogue-Check: no-match → vendored Herald-internal (<code>docs/catalogue-checks/HRD-099-commons-auth.md</code>).
<code>bherald (CLI)</code>	✅ scaffolded (Wave 2 / HRD-095)	Build Herald CLI-only binary. 3 §43 stubs (HRD-035/041/045).
<code>rherald (CLI)</code>	✅ scaffolded (Wave 2 / HRD-096)	Release Herald CLI-only binary. 3 §43 stubs (HRD-031/032/045).
<code>iherald (CLI)</code>	✅ scaffolded (Wave 2 / HRD-097)	Incident Herald binary serving <code>:24794</code> via <code>cli.ServeCmd. /v1/webhooks/page</code> returns honest 501 + HRD-024.
<code>scherald (CLI)</code>	✅ scaffolded (Wave 2 / HRD-097)	Scheduled-audit Herald CLI-only binary. 1 §43 stub (HRD-047).

Operations

- **Repo hygiene:** main only; four-mirror fan-out proven across spec + Foundation M1/M2/M3 + anti-bluff infra + §107 covenant landing.
- **Constitution inheritance gate: 15 PASS / 0 FAIL** (was 12; I8a/b/c added 2026-05-20 for §107 covenant).
- **Anti-bluff audit: 16 PASS / 0 FAIL / 1 SKIP** (was 14; HRD-011 telebot.v3 third-party SKIP-with-reason per §11.4.74; I8 paired meta-test included).
- **CodeGraph validate:** 7 PASS / 0 FAIL / 2 SKIP (HRD-091 documented submodule-traversal gap).
- **E2E bluff-hunt against real services: 41 PASS / 0 FAIL / 5 SKIP** (was 33; Wave 3a added 8 new PASSES: E35/E36 auth gate × 2 flavors + E43/E44 cherald compliance live + E46/E47 sherard safety_state live; updated E28/E29 from "501 stub" → "401 JWT-gated" to reflect Wave 3a route swap). SKIPS are E17/E18/E34 (Telegram live, claude live, full inbound — requires operator credentials) + E45 (cross-binary cherald-reads-pherard-writes — Wave 3b once Runner lands) + E48 (sherard destructive-op trigger — HRD-033 body not implemented). Real Gin server, live `/v1/*` endpoints, real Postgres + Redis containers via `containers/` submodule, M2 RLS integration tests + HRD-010 storage tests against the live stack, SIGTERM-graceful-shutdowns. Paired §1.1 mutation gates: `tests/test_wave2_mutation_meta.sh` (4/4 PASS — Wave 2 invariants) + `tests/test_wave3_mutation_meta.sh` (3/3 PASS — Wave 3a invariants M1+M5+post-flight, M6 SKIP). This is the canonical §107 evidence per Herald §107.
- **Submodules:** 12 vendored — 11 Helix-stack capability modules under `submodules/` (auth, background, cache, config, database, eventbus, middleware, observability, recovery, plus Models + Concurrency added during HRD-010) referenced via `replace` directives in consuming `go.mod` files, NOT via `go.work`; plus `containers/` (runtime auto-detection + on-demand container boot,

consumed directly by Foundation tests + pherald doctor). All 12 carry the §11.4 anti-bluff anchor.

- **Build / test toolchain:** Go 1.25+ (verified on 1.26 per CLAUDE.md), Pandoc 3.9, WeasyPrint 66, Podman 5.8 — verified locally.

Risk surface

- **Submodule-catalogue discovery (operator mandate 2026-05-20).** Every new HRD MUST carry a Catalogue-Check: `reuse|extend|no-match <org/>repo>@<sha>` line per Universal §11.4.74. New HRDs (HRD-029..056 §43 catalogue) currently default to TBD pending operator review.
- **Versioning discipline (operator mandate 2026-05-20).** Minor changes bump secondary version, major bump primary. Currently at V3 r7→r8 within the V3 series; a full V4 rewrite is reserved for a primary-version jump.
- **Live channel integrations untested at runtime.** HRD-011 (Telegram telebot live) requires operator-supplied credentials per `docs/CONTINUATION.md`. HRD-012 (Claude Code `claude --resume` live) is closed with live PASS evidence captured. The §107 covenant blocks claims of "done" on these until end-user evidence is captured.
- **HRD-008 operator-side quickstart not yet validated end-to-end on a fresh laptop.** The compose stack works in dev; operator hardware validation is the next concrete §107 evidence point.
- **No CI yet.** No GitHub Actions / GitLab CI / etc. Operator runs `scripts/e2e_bluff_hunt.sh` manually for now — that script IS the local CI surrogate.

Closed risks

- **Production RLS-bypass bug (closed 2026-05-20 during HRD-010 step 3, commit 5c1c022).** Until that commit, `commons_infra.WithTenantContext` set the tenant GUC but `ROW LEVEL SECURITY` was not FORCED on the targeted tables — and the bootstrap PG user used in tests was a superuser, which bypasses RLS regardless. The E14 falsifiability test (real Postgres, cross-tenant SELECT must return EXACTLY 0 rows from the other tenant) caught the bluff before any live deployment. Fix: switched the application connection to a non-superuser role + ran `ALTER TABLE ... FORCE ROW LEVEL SECURITY` in migration 000003. The §107 covenant directly prevented this from shipping; the lesson is "EXACT-N row count, never >=N" for any isolation test.

Recent activity

Wave 3a — substrate + 2 live REST routes (closed 2026-05-21)

8-step workstream commits `dbbea95` . . this shipped:

- `commons_auth/ JWT verifier scaffold` (HRD-099, renumbered from HRD-093; commit `dbbea95`) — new top-level Go module: HMAC (HS256) + JWKS (RS256/ES256) hybrid verifier with Redis-cached keys + in-memory fallback; Gin middleware (`GinMiddleware`) that gates every request behind the verifier and sets `ContextKeyClaims` on success; `claims.go` helpers (`TenantFromClaims`, `SubjectFromClaims`); `verifier_test.go` exercises both modes incl. expired-token + wrong-signature paths. Catalogue-Check: no-match → vendor Herald-internal (`docs/catalogue-checks/HRD-099-commons-auth.md`).

- **events_processed migration** (commit 3f8daed) — `commons_storage/migrations/000012_events_processed.{up,down}.sql`; RLS-enforced 30-day TTL inbound idempotency archive per V3 §32.2.
- **ConstitutionStore.ListQuery extension** (commit 6276437) — Since/Until/Offset fields added; both Memory + Postgres backends honour them.
- **cherald/internal/compliance/ Handler + tests** (commit a2ef831) — paginated + filter-aware `/v1/compliance` handler; tenant scope from JWT claims; defaults `page=1`, `page_size=50` (max 200).
- **cherald wiring + route swap** (HRD-028 close-out, commit bfca740) — `cherald/cmd/cherald/main.go` builds the Verifier from env, instantiates the ConstitutionStore (Postgres if `HERALD_PG_DSN` else Memory), and wires `commons_auth.GinMiddleware + compliance.Handler` via `cli.ServeOpts.Middleware`. The 501-stub for `/v1/compliance` is gone.
- **sherald/internal/safety/ Aggregator + Handler + sampler** (commit 1016b96) — process-local Aggregator tracks `lastMemPercent + lastDestructiveOp + uptime`; `StartMemSampler` goroutine ticks every 15s; Handler returns the snapshot as JSON; tests cover the Aggregator surface.
- **sherald wiring + sampler start** (HRD-098 close-out, commit bfa82bc) — `sherald/cmd/sherald/main.go` instantiates the Aggregator + starts the sampler under a SIGTERM-bound context; wires the Verifier + Handler via `cli.ServeOpts.Middleware`. The 501-stub for `/v1/safety_state` is gone.
- **e2e + mutation gate + docs sync (commit 0cc6fad)** — 8 new e2e invariants (E35/E36/E43/E44/E46/E47 PASS; E45/E48 SKIP-with-reason); E28/E29 updated from "501 stub" to "401 JWT-gated"; e2e header set process-wide `HERALD_AUTH_MODE + CODEGRAPH_ALLOW_UNSAFE_NODE`; new paired §1.1 mutation gate tests / `test_wave3_mutation_meta.sh` (3/3 PASS); atomic Issues→Fixed for HRD-028, HRD-098, HRD-099 (`commons_auth`, renumbered from HRD-093 in this r11 cleanup); `Status.md` r9 → r10 → r11; `Issues.md` r8 → r9 → r10; `Fixed.md` r7 → r8 → r9.

Anti-bluff battery sign-off (this commit): inheritance gate 15/15 PASS; meta-tests (I6, I8, Wave2, Wave3) all green; `audit_antibluff.sh` 16 PASS / 0 FAIL / 1 SKIP (`commons_auth` not added to the audit's submodules/ walk — workspace-internal module by design); `codegraph_validate.sh` 7 PASS / 0 FAIL / 2 SKIP (with `CODEGRAPH_ALLOW_UNSAFE_NODE=1` for Node 25 environment); `e2e_bluff_hunt.sh` 41 PASS / 0 FAIL / 5 SKIP.

Wave 2 — flavor-scaffold sweep (closed 2026-05-21)

15-step workflow commits 7e0a614 . . eef606b (with parallel logo branding 24b96f2) shipped:

- **Shared scaffold** — new `commons/cli/` package: `NewRootCmd` (steps 2), `VersionCmd` (step 2), `routes.go + healthz/readyz/metrics` handlers (step 3), `ServeCmd` with `Middleware-hook + graceful shutdown` (step 4), `StubCmd` honest 501-style stubs (step 1). Vendored Herald-internal per §11.4.74 catalogue-check no-match → see `docs/catalogue-checks/HRD-092-commons-cli.md`.
- **pherald refactor** — `main + stubs` migrated to consume `commons/cli/` (step 5); `pherald serve` consumes `cli.ServeCmd` with `RequestIDMiddleware` hook (step 6, commit 31562cf).
- **6 new flavor binaries** — `sherald` serving :24793 (step 7, HRD-093), `cherald` serving :24792 (step 8, HRD-094), `iherald` serving :24794 (step 9, HRD-097), `bherald` CLI-only (step 10, HRD-095), `rherald` CLI-only (step 11, HRD-096), `scherald` CLI-only (step 12, HRD-097). Each ships honest 501-style §43 command stubs with HRD pointers.

- **Anti-bluff additions** — `scripts/e2e_bluff_hunt.sh` grew from 18 → 33 invariants covering all 6 new binaries (version --json shape, healthz binding, /v1/safety_state + /v1/compliance + /v1/webhooks/page 501 honesty, sherald SIGTERM graceful-shutdown, §43 stub exit-code, pherald regression sentinel). New paired mutation gate tests `tests/test_wave2_mutation_meta.sh` (4/4 PASS) proves the new invariants catch their claimed regressions.
- **Spec capture** — V3 r7 → r8 added §6.3 Branding extension (per-flavor fields), §18.0 Wave 2 catalogue, §41 REST route additions, §44.M Wave 2 milestone, §43.2 catalogue-check addendum (step 14, commit `ee f606b`).
- **Atomic Issues→Fixed** — HRD-092..097 closed in r7 of Fixed.md; HRD-098 opened in r8 of Issues.md as Wave 3+ deferred (sherald /v1/safety_state live impl).

Anti-bluff battery sign-off (this commit): inheritance gate 15/15 PASS; meta-tests (I6, I8, Wave2) all green; `audit_antibluff.sh` 16 PASS / 0 FAIL / 1 SKIP; `codegraph_validate.sh` 7 PASS / 0 FAIL / 2 SKIP; `e2e_bluff_hunt.sh` 33 PASS / 0 FAIL / 3 SKIP.